Vishal Indivar Kandala

VishalKandala | vishalkandala | vishalkandala.me
vishalkandala@tamu.edu | +1 (979) 250-7375

Research Profile

Ph.D. student in Mechanical Engineering specializing in computational fluid dynamics, turbulence modeling, Eulerian–Lagrangian methods, and high-performance scientific computing. My dissertation research focuses on developing PICurv, a parallel Particle-in-Cell solver on curvilinear grids for Filtered Density Function / Velocity Filtered Density Function turbulence closure modeling. My broader research experience includes biomedical CFD simulations of LVAD outflow graft hemodynamics, physics-informed modeling of cardiac pressure-volume relationships, and scientific software development for complex flow problems.

Education

2021 - present **Ph.D., Mechanical Engineering** at Texas A&M University — College Station, TX
Advisor: Dr. Iman Borazjani
Dissertation: Data-Driven Turbulence Closure Models using Filtered Density Function Methods

2019 - 2021 **M.Eng., Aerospace Engineering** at Texas A&M University (GPA: 3.8/4.0)
College Station, TX

2014 - 2018 **B.Tech., Mechanical Engineering** at VIT University (GPA: 8.0/10.0)
Chennai, India

Research Interests

Primary Focus	Turbulence modeling, Data-driven closure development, Filtered Density Function (FDF) methods, Large Eddy Simulation (LES)
Computational Methods	High-performance computing, Eulerian-Lagrangian methods, Immersed boundary methods, Parallel computing, Numerical methods for PDEs
Application Areas	Turbulent scalar transport, Combustion modeling, Fluid-structure interaction, Biomedical flows, Particle-laden flows
Emerging Interests	Physics-informed machine learning, Scientific machine learning for turbulence, Hybrid numerical-ML approaches

Research Experience

Graduate Research Assistant — Texas A&M University May 2021 - Present

Scientific Computing and Biofluids Lab, Advisor: Dr. Iman Borazjani

PICurv: Eulerian-Lagrangian Solver for Turbulence Closure Modeling. Jan 2024 - Present

- Developing a high-fidelity solver for scalar Filtered Density Function (FDF) transport within the CURVIB immersed boundary framework.
- Implementing Eulerian-Lagrangian coupling and solving stochastic particle methods (Langevin equation) to model sub-grid scalar fluctuations in turbulent flows.
- Building a modular, well-documented, research codebase with a focus on robust software engineering for the scientific community.
- Optimizing parallel scalability to handle millions of particles across 1,000+ cores using C/C++, MPI, and PETSc.

– This work establishes the foundation for a future velocity-scalar FDF (VFDF) framework, enabling a new class of data-driven turbulence closure models.

Biomedical Computational Mechanics Projects May 2021 - Dec 2023

Left Ventricular Assist Device (LVAD) Flow Analysis:

— Conducted high-resolution, unsteady fluid-structure interaction simulations of pulsatile blood flow within patient-specific LVAD outflow graft geometries.

— Analyzed key hemodynamic metrics including **wall shear stress, vorticity, and regions of flow stasis** to quantify the risk of thrombosis (blood clotting) and hemolysis (blood cell damage).

— Utilized the Curvilinear Immersed Boundary (CURVIB) method to accurately capture the interaction between blood flow and the flexible arterial walls.

— Presented a detailed analysis of the outflow graft hemodynamics at the 76th Annual APS Division of Fluid Dynamics meeting.

Cardiac Pressure-Volume Modeling (EDPVR):

— In a cross-disciplinary collaboration with the Cardiovascular and Computational Biomechanics Lab, developed a novel framework for modeling right ventricular pressure-volume dynamics.

— Designed and implemented a **physics-informed regression model** that extends the established Klotz single-beat method to the anatomically complex right ventricle.

— Generated training data by performing finite element simulations (Abaqus) on a large cohort of biventricular geometries reconstructed from high-resolution MRI scans.

— Demonstrated a **40-60% reduction in prediction error** (nRMSE reduced from 22.7% to 12.3%) by introducing a geometric correction factor, enabling more accurate, non-invasive cardiac assessment. Manuscript in preparation for *Nature Scientific Reports*.

Undergraduate Research Assistant — VIT University Jun 2017 - May 2018

Advisor: Dr. Harish Rajan

Hypersonic Reentry Vehicle Design Optimization

– Conducted a design optimization study for a modified double-spiked reentry capsule using ANSYS Fluent to mitigate aerothermal loads.

– Performed parameter sweeps varying aerospike frustum geometry focused on drag and surface temperature reduction.

– Authored a technical report on the findings, demonstrating a systematic approach to design space exploration through computational simulation.

Publications & Presentations

Conference Presentations

Kandala, V.I. and Borazjani, I. 2025

"A Parallel Particle-In-Cell (PIC) Solver on Curvilinear Grids for Turbulent Flow Simulation," *78th Annual APS Division of Fluid Dynamics*, Houston, TX.[Schedule]

Kandala, V.I. and Borazjani, I. 2023

"High Resolution Numerical Simulations of LVAD Outflow Graft Haemodynamics," *76th Annual APS Division of Fluid Dynamics*, Washington, DC. [Abstract] [Presentation]

Posters

Naeini, V., **Kandala, V.**, et al. 2023

"On the non-invasive estimation of end diastolic pressure volume relationship in the right ventricle," *BMES Annual Meeting*, Seattle, WA. [Abstract]

Kandala, V.I. and Borazjani, I. 2022

"High Performance Computing for Biofluids," *High Performance Research Computing Symposium*, Texas A&M University. [Poster] [Event]

Manuscripts in Preparation

Naeini, V., **Kandala, V.I**, et al.

"In-silico-generated universal end-diastolic pressure-volume relationship for the right ventricle," *in preparation for Nature Scientific Reports.*

Awards & Honors

2022 **Chad Patel Scholarship**, American Society of Indian Engineers

2019, 2020 **Aerospace Academic Excellence Fellowship**, Texas A&M University

Teaching Experience

Teaching Assistant — Texas A&M University Mar 2020 - May 2025

STAT 201: Statistical Methods Spring 2020 - Spring 2023

- Instructed 200+ students from diverse academic backgrounds in statistical inference
- Developed materials emphasizing practical applications and conceptual understanding

MEEN 461: Heat Transfer Fall 2023

- Taught graduate students and senior undergraduates from ME, AE, and ChemE programs
- Created assignments, projects, and exams; delivered lectures; conducted tutorials using MATLAB and Python

MEEN 464: Heat Transfer Laboratory Fall 2024, Spring 2025

- Supervised small teams (4-5 students) conducting heat transfer experiments
- Developed lab manuals and experimental protocols; managed laboratory equipment

Service & Leadership

Judge — Texas Science Fair 2023

Evaluated student research projects and provided feedback to young researchers

Department Representative — TAMU Graduate and Professional Student Government 2

Represented mechanical engineering graduate students in university governance

Contributing Writer — Ingenium Blog, Texas A&M Engineering 2020 - 2022

Wrote articles about graduate student experiences and research for university engineering blog

Co-Founder & Head of Aerodynamics — Team Aviators International 2016 - 2017

- Founded first SAE Aero Design team at VIT University; led 5-member aerodynamics sub-team
- Secured $5,000+ in funding through workshops and sponsorships

Lead Braking Engineer — Team Saksham (Baja SAE India) 2015 - 2016

- Designed and fabricated braking system; achieved 5th place nationally (100+ teams)
- Contributed to fundraising efforts securing $3,000+ for components and travel

Research Software and Computational Projects

PICurv — Documentation

High-fidelity Eulerian-Lagrangian solver for particle-laden turbulent flows on curvilinear grids. Features modular design, comprehensive Doxygen documentation, automated test cases, and parallel scalability.

Cantera-Radcal GitHub

Extension of Cantera chemical kinetics package with radiation modeling (RADCAL) and higher-order diffusion. Docker implementation for reproducibility.

Parallel Diffusion Equation Solver GitHub

2D diffusion equation solver using finite difference method with PETSc library's C interface. Features Krylov subspace methods for efficient parallel solution.

Py-SIMPLE GitHub

2D finite volume method solver for incompressible, steady-state flows written in Python with NumPy. Implements SIMPLE algorithm for pressure-velocity coupling.

Parallel Artificial Compressibility Solver GitHub

High-performance CFD solver for 2D steady-state incompressible Navier-Stokes equations on generalized curvilinear grids using artificial Compressibility method.

1D Composite Diffusion Solver GitHub

Simulates 1D heat conduction in spherical bodies with multiple layers of different material compositions and varying thicknesses in the radial direction.

Technical Skills

Programming & HPC	C/C++, Python, MATLAB, MPI, OpenMP, PETSc, CUDA (exposure)
CFD & Simulation	ANSYS Fluent, OpenFOAM, STAR-CCM+ (exposure), Converge (exposure), Cantera
Numerical Methods	Finite Volume Method, Finite Element Method, Immersed Boundary Methods, Spectral Methods, Time Integration (RK, fractional-step), Iterative Solvers (GMRES, Krylov)
Meshing & Pre-processing	ICEM CFD, Pointwise, Gmsh, ANSYS Meshing
Post-processing	Tecplot, ParaView, EnSight, Matplotlib
CAD	SolidWorks, Creo, SpaceClaim
HPC Systems	National supercomputing resources, cluster management, job scheduling
Version Control	Git, GitHub

Relevant Coursework

CFD & Turbulence Modeling	Turbulence Modeling (AERO 670), Turbulence Processes (OCEN 640), Computational Fluid Dynamics (MEEN 689)
Propulsion & High-Speed Flows	High-Speed Combustion & Propulsion (AERO 641), Aerothermochemistry (AERO 676), Advanced Aerodynamics (AERO 601)
Scientific & Parallel Computing	Scientific Machine Learning (ECEN 689), Parallel Computing (CSCE 689), Advanced Computer-Aided Engineering (MEEN 632)
Numerical Methods & Analysis	Numerical Heat Transfer & Fluid Flow (MEEN 644), Numerical Methods in Differential Equations (MATH 610), Numerical Analysis (AERO 689)
Fundamental Mechanics	Theory of Fluid Mechanics (AERO 602), Fluid Mechanics (MEEN 621), Introduction to Random Dynamical Systems (AERO 630)

Professional Memberships

American Physical Society (APS) – Division of Fluid Dynamics

Professional Experience

Product Development Intern — EPC Electrical Pvt. Ltd. Jan 2018 - Aug 2018

- Co-led the end-to-end design, fabrication, and deployment of a novel, ultra-low-cost CNC machine, achieving a time-to-market of under 90 days.
- Reduced customer operating costs by 60% by implementing the GRBL open-source firmware and control platform.

R&D Engineering Intern — BHEL Jun 2017 - Aug 2017

- Selected as one of 20 interns from a pool of 300+ applicants to contribute to R&D at India's largest energy equipment producer.
- Developed a MATLAB application to compute heat transfer coefficients in industrial condensers, eliminating commercial software licenses and saving an estimated $2,500 annually.

Independent Projects

Cantera-Radcal: Chemical Kinetics with Radiation Mar 2020 - Jan 2021

- Extended the open-source Cantera package to include radiation modeling and higher-order reverse diffusion as part of my master's thesis project.
- Integrated the modified package with a compressible OpenFOAM solver to simulate non-premixed flames, a critical step for high-speed combustion applications.

Foundational CFD Solver Implementations Coursework & Self-Study

- Developed a portfolio of solvers from first principles to build a fundamental understanding of numerical methods and solver architecture.
  - **Parallel Diffusion Solver:** A 2D Finite Difference solver in C using the PETSc library for parallel execution and scalability analysis. [GitHub]
  - **Artificial Compressibility Solver:** A 2D Navier-Stokes solver for backward-facing step flow using an explicit Runge-Kutta time-marching scheme on a stretched grid. [GitHub]
  - **SIMPLE Algorithm Solver:** A 2D Finite Volume solver in Python implementing the SIMPLE algorithm on a staggered grid for channel and lid-driven cavity flows. [GitHub]

References

Dr. Iman Borazjani *Ph.D. Advisor*
Professor, J. Mike Walker '66 Department of Mechanical Engineering
Texas A&M University — Email: iman@tamu.edu

Dr. Reza Avazmohammadi *Research Collaborator*
Associate Professor, Department of Biomedical Engineering
Texas A&M University — Email: rezaavaz@tamu.edu

Dr. Harish Rajan *Undergraduate Research Advisor*
Associate Professor, School of Mechanical Engineering
Vellore Institute of Technology (VIT), Chennai — Email: harish.r@vit.ac.in